AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2016

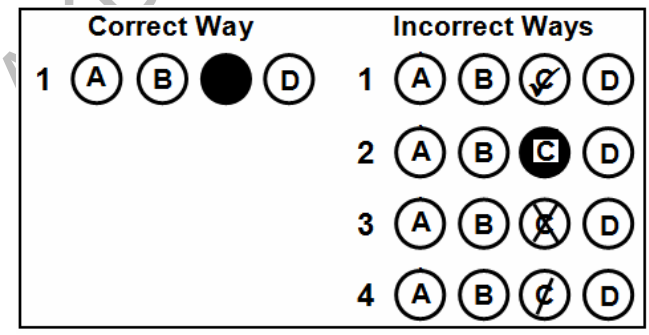
Computer Science Paper I

Time: 35 minutes Marks: 25

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

1. All of the following are harmful effects of a computer virus EXCEPT that it

   A. corrupts data files.
   B. erases the contents of a folder.
   C. physically destroys the infected disk.
   D. lies dormant for a specified time and then becomes active.

2. Which of the following are the benefits of using robots in industry?

   I. Flexibility
   II. Perform repetitive tasks without being bored
   III. Great strength
   IV. High level of precision

   A. I, II and III
   B. I, III and IV
   C. I, II and IV
   D. II, III and IV

3. Using ideas and creation of other people without their prior permission is called

   A. plagiarism.
   B. data mining.
   C. downloading.
   D. copying and pasting.

4. During internet access, it is legal to

   A. spread computer viruses.
   B. download and use copyright material as your own.
   C. hide one's personal information on the web.
   D. use each other's computers without permission.

5. Computers are playing an important role in reducing pollution by

   A. deskilling people.
   B. reducing paper work.
   C. consuming less time.
   D. using anti-virus software.

6. Computer programs can consist of all of the following elements EXCEPT

   A. comments.
   B. looping structures.
   C. flow chart symbols.
   D. variables and constants.

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7. The input of the compilation process before running the program is

   A. debug code.
   B. object code.
   C. source code.
   D. pseudo code.

8. All of the following are basic steps for writing an algorithm EXCEPT

   A. supplying input.
   B. generating output.
   C. performing debugging.
   D. performing mathematical operations.

9. Which of the following flow chart symbols is used for IF THEN statement?

   A. 

   B. 

   C. 

   D. 

10. Which of the following commands is used to return from GW BASIC to DOS prompt?

    A. END
    B. STOP
    C. CLEAR
    D. SYSTEM

11. In GW BASIC, key F3 is used to

    A. run the program.
    B. load saved program from disk.
    C. list the program lines on screen.
    D. save programs from memory to disk.

12. In GW BASIC, variables beginning with FN are referred to as

    A. built-in function.
    B. built-in subroutine.
    C. user-defined function.
    D. user-defined subroutine.

13. Consider the given GW BASIC program.

```
10 IF A >= 0 THEN B = A + 2 ELSE B = A – 2
20 PRINT B
```

If A has the value 3, which of the following values of B will print at line # 20?

A. –2
B. 1
C. 2
D. 5

14. Which of the following is an example of nested loop in a GW BASIC program?

A. GOTO within IF THEN
B. GOTO within FOR NEXT
C. WHILE WEND within FOR NEXT
D. IF THEN ELSE within WHILE WEND

15. The Body Mass Index (BMI) categorises a person according to the following table.

BMI	Category
Greater than 25	Over-weight
Between 25 and 19	Normal
Less than 10	Under-weight

If a program is written in GW BASIC, which of the following statements is most likely to be used to display a category for a BMI?

A. Goto
B. For..Next
C. While-Wend
D. If..Else..Endif

16. Consider the following GW BASIC program.

```
For i = ? to? step?
Print i
Next i
```

What values will you assign to the loop to produce the following series of numbers?

100, 98, 96, 94, ……. 6, 4, 2

A. For i = 2 to 100 Step 2
B. For i = 100 to 2 Step 2
C. For i = 2 to 100 Step –2
D. For i = 100 to 2 Step –2

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17. Consider the following GW BASIC program.

```
10 FOR A = 1 TO 5 STEP 2
20 PRINT A
30 NEXT A
```

The number of repetitions in the program is

A. 1
B. 2
C. 3
D. 5

18. All of the following are valid about arrays in GW BASIC EXCEPT

A. an array subscript may have a negative value.
B. an array can generate "subscript out of range" error.
C. the maximum number of dimensions for an array is 255.
D. an array variable name can be used without a DIM statement.

19. The output of the GW BASIC statement PRINT ABS(– 9.5) – INT(– 9.5) will be

A. –19.5
B. –18.5
C. 18.5
D. 19.5

20. Which of the following will draw a circle of radius 50 in GW BASIC?

A. CIRCLE 50, (100, 100)
B. CIRCLE (100, 100), 50
C. CIRCLE (50, 100, 100)
D. CIRCLE (100, 100, 50)

21. The output of GW BASIC statement PRINT “2” + “101” will be

A. 103
B. 301
C. 2101
D. 1012

22. In GW BASIC, the CORRECT way for writing the expression $\sqrt[3]{64}$ is

A. (64) ^ 1/3
B. (64) ^ (1/3)
C. SQR (64) ^ 1/3
D. SQR (64) ^ (1/3)

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23. In GW BASIC, two or more statements are separated by

A. ;
B. :
C. ,
D. #

24. The complement of the Boolean expression $\overline{A} + B$ will be

A. $\overline{A}.B$
B. $A.\overline{B}$
C. $A + \overline{B}$
D. $\overline{A} + \overline{B}$

25. Consider the given truth table.

A	$\overline{A}$	$\overline{\overline{A}}$
1	X	Y

Which of the following are the values of X and Y respectively?

A. 1 and 0
B. 1 and –1
C. –1 and 1
D. 0 and 1

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END OF PAPER

Please use this page for rough work

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Please use this page for rough work

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